

HabitFlow API Specification

Team-404

Version 1.1

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Endpoints marked with an Authorization parameter require the header Authorization: Bearer <token> from /api/auth/login. Every authenticated request only sees data owned by that user.

V. Task Organization (api/priorities, api/labels)

Task organization covers Priority Levels and Labels/Tags (WBS 5.3). This section defines priorities and labels. Assigning them to a task is done in the Task API (III):

Priority: set with Add Task (III.C) and changed with Update Task (III.D) or Update Task Priority (III.H).

Labels: attached to or removed from a task with Set Task Labels (III.I). A task's current labels are returned by List Tasks (III.A) and Get Task (III.B).

A. List Priorities

API Endpoint	/api/priorities	
method	GET	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
<i>Lists the fixed priority levels (tbl_priority_level) that can be assigned to a task, ordered by sortOrder. Priority 4 is the default for new tasks.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
[]	array	List of priority levels.
{}.priorityId	integer	Priority ID.
{}.name	string	Priority name (max 50 chars).
{}.sortOrder	integer	Display order.
Sample Response		
[{"priorityId": 1, "name": "Priority 1", "sortOrder": 1}, {"priorityId": 2, "name": "Priority 2", "sortOrder": 2}, {"priorityId": 3, "name": "Priority 3", "sortOrder": 3}, {"priorityId": 4, "name": "Priority 4", "sortOrder": 4}]		
Fail		
Status Code	401	
Response Type	json	
Response Data	Type	Description

error	string	Authentication error.
Sample Response		
{"error": "Unauthorized"}		

B. List Labels

API Endpoint	/api/labels	
method	GET	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
<i>Lists all labels created by the authenticated user.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
[]	array	List of labels.
[] .labelId	integer	Label ID.
[] .name	string	Label name.
[] .color	string/null	Label color (hex, e.g. #2563EB).
Sample Response		
[{"labelId": 3, "name": "Study", "color": "#2563EB"}, {"labelId": 4, "name": "Habit", "color": "#16A34A"}]		
Fail		
Status Code	401	
Response Type	json	
Response Data	Type	Description
error	string	Authentication error.
Sample Response		
{"error": "Unauthorized"}		

C. Create Label

API Endpoint	/api/labels	
method	POST	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
name	string	Label name (required, max 50 chars).
color	string/null	Optional color value (hex, e.g. #F59E0B).

Creates a new label for the authenticated user. Attach it to tasks with Set Task Labels (III.I).

Success

Status Code	200	
Response Type	json	
Response Data	Type	Description
labelId	integer	Label ID.
name	string	Label name.
color	string/null	Label color (hex).

Sample Response

```
{"labelId": 5, "name": "Errands", "color": "#F59E0B"}
```

Fail

Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.

Sample Response

```
{"error": "Label name is required"}
```

D. Update Label

API Endpoint	/api/labels/{labelId}	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
labelId	integer	Existing label ID (path parameter).
name	string	Updated label name (max 50 chars).
color	string/null	Updated color (hex). null removes the color.
<i>Updates several details of a label at once (e.g. from an Edit Label form). Fields not sent are left unchanged. To change a single field, use Rename Label (V.E) or Update Label Color (V.F). The change shows on every task that uses the label.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
labelId	integer	Label ID.
message	string	Success confirmation.
Sample Response		

{"labelId": 5, "message": "Label updated successfully"}		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{"error": "Label name cannot be empty"}		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Label not found or not accessible.
Sample Response		
{"error": "Invalid updating label."}		

E. Rename Label

API Endpoint	/api/labels/{labelId}/name	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
labelId	integer	Existing label ID (path parameter).
name	string	New label name (required, max 50 chars).
<i>Changes only the name of a label owned by the authenticated user.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
labelId	integer	Label ID.
name	string	Updated label name.
message	string	Success confirmation.
Sample Response		
{"labelId": 5, "name": "Chores", "message": "Label renamed successfully"}		
Fail		
Status Code	400	

Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{"error": "Label name is required"}		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Label not found or not accessible.
Sample Response		
{"error": "Label not found"}		

F. Update Label Color

API Endpoint	/api/labels/{labelId}/color	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
labelId	integer	Existing label ID (path parameter).
color	string/null	New color (hex, e.g. #DC2626). null removes the color.
<i>Changes only the color of a label owned by the authenticated user.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
labelId	integer	Label ID.
color	string/null	Updated color.
message	string	Success confirmation.
Sample Response		
{"labelId": 5, "color": "#DC2626", "message": "Label color updated successfully"}		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.

Sample Response		
{ "error": "Invalid color format" }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Label not found or not accessible.
Sample Response		
{ "error": "Label not found" }		

G. Delete Label

API Endpoint	/api/labels/{labelId}	
method	DELETE	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
labelId	integer	ID of the label to delete (path parameter).
<i>Permanently deletes a label and removes it from every task that uses it (its tbl_task_label rows). The tasks themselves are not deleted.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
message	string	Label deletion confirmation.
Sample Response		
{ "message": "Label deleted successfully" }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Label not found or not accessible.
Sample Response		
{ "error": "Label not found" }		